

Q4 2024 Earnings Call

Company Participants

- Bren Higgins, Executive Vice President and Chief Financial Officer
- Kevin Kessel, Vice President, Investor Relations
- Rick Wallace, President and Chief Executive Officer

Other Participants

- Atif Malik, Citigroup
- Blayne Curtis, Jefferies, LLC
- Brian Chin, Stifel
- C. J. Muse, Cantor Fitzgerald
- Chris Caso, Wolfe Research
- Harlan Sur, J.P. Morgan Chase & Co.
- Joe Quatrochi, Wells Fargo
- Joseph Moore, Morgan Stanley
- Krish Sankar, TD Cowen
- Srini Pajjuri, Raymond James
- Timothy Arcuri, UBS Securities
- Toshiya Hari, Goldman Sachs & Co.
- Vivek Arya, Banc of America Merrill Lynch

Presentation

Operator

Good afternoon. My name is David and I'll be your conference operator today. At this time, I would like to welcome everyone to the KLA Corporation June Quarter 2024 Earnings Conference Call and Webcast.

All participants have been placed on a listen-only mode to prevent any background noise. After the speakers' remarks, there will be a question-and-answer session. (Operator Instructions). Please limit yourself to one question and one follow-up. (Operator Instructions). Thank you.

I will now turn the call over to Kevin Kessel, Vice President of Investor Relations and Market Analytics. Please go ahead.

Kevin Kessel {BIO 16743902 <GO>}

Welcome to our earnings call to discuss the June 2024 results and the September quarter outlook. I am joined by our CEO, Rick Wallace; and our CFO, Bren Higgins. We will discuss today's results released after the market close and available at ir.kla.com, along with supplemental materials.

Today's discussion and metrics are presented on a non-GAAP financial basis unless otherwise specified. All full year references are to calendar years. The earnings materials contain a detailed reconciliation of GAAP to non-GAAP results. KLA's IR website also contains future investor events, presentations, corporate governance information, and links to our SEC filings, including our most recent Annual Report and quarterly reports on forms 10-K and 10-Q.

Our comments today are subject to risks and uncertainties reflected in the disclosure of Risk Factors in our SEC filings. Any forward-looking statements, including those we make on the call today, are also subject to those risks, and KLA cannot guarantee those forward-looking statements will come true. Our actual results may differ significantly from those projected in our forward-looking statements.

Rick will begin the call with some introductory comments, followed by Bren with additional financial highlights, including our outlook.

Let me turn the call over to our CEO, Rick Wallace. Rick?

Rick Wallace {BIO 3540990 <GO>}

Thank you, Kevin.

I will cover a few business highlights for this quarter. KLA's June quarter revenue, gross margin, and EPS, were all above their respective guidance midpoints. The quarter played out mostly as expected, with strong customer demand and solid company execution. We continue to see signs of a strengthening market environment for customers at the leading edge and remain confident in our plan for steady improvement throughout the remainder of this calendar year and into 2025.

This quarter, KLA results returned to sequential and year-over-year revenue growth, demonstrating an improving industry environment. In Foundry & Logic, a continuation of scaling and incorporation of new technologies, and slowly rising capital intensity, continues to be a long-term tailwind. In Memory, technology development investments, including AI, high-bandwidth memory, and the improving supply-demand environment, support an expected return to growth for Memory markets in 2025.

The June quarter also marked another quarter of strong performance from our portfolio of Broadband Plasma, or BBP products, which we also refer to as Gen4 and Gen5. This year marks 4 years of innovation for our BBP patterned wafer inspection business. We continue to see significant differentiation, keeping this product line at the forefront of defect discovery for the semiconductor industry. We expect this product family momentum to continue delivering strong relative growth performance in calendar year 2024.

AI continues to be a driver and enabler for KLA's business. AI adoption is driving higher volume wafer manufacturing, more complex designs, larger die and chip size, and growing advanced packaging demands which support an increase in process control intensity. Last quarter, we estimated that the KLA opportunity would reach approximately \$400 million in annualized revenue exiting calendar 2024. Today, we're raising that estimate to more than \$500 million in total advanced packaging revenue across the entire KLA portfolio on the momentum we're experiencing.

Additionally, AI is enabling KLA to further differentiate our products and make continuous improvement in the performance of our systems. KLA's service business grew to \$614 million in the June quarter, up 4% sequentially and 14% year-over-year. Utilization rates of existing installed capacity are steadily rising across all business segments as end market demand improves.

Finally, quarterly free cash flow was \$832 million. The last 12-month free cash flow was \$3 billion, with free cash flow margin of 31% over the same period.

KLA results continue to demonstrate our sustained process control leadership and the success of our broad portfolio and specific product strategies. In this industry environment, KLA will continue to focus on supporting customer requirements, executing on product roadmaps, and preparing for growth at the leading edge.

I'll now hand the call over to Bren to go through the financial highlights and outlook.

Bren Higgins {BIO 18108981 <GO>}

Thanks, Rick.

KLA's quarterly results demonstrated our market leadership combined with the consistent execution of our global team. KLA continues to show resourcefulness and the ability to adapt to meet customers' changing requirements.

Quarterly revenue was \$2.57 billion, above the guidance net midpoint of \$2.5 billion. Non-GAAP diluted EPS was \$6.60 and GAAP diluted EPS was \$6.18, both above their respective guidance midpoints.

Gross margin was 62.5%, at the upper-end of the guidance range as a richer product mix than modeled and higher revenue volume drove upside to guidance. Operating expenses were \$553 million. Operating expenses were comprised of \$324 million in R&D and \$229 million in SG&A. Operating margin was 41%.

Other income and expense net was a \$32 million expense, and the quarterly effective tax rate was 12.6%. Quarterly non-GAAP net income was \$893 million. GAAP net income was \$836 million. Cash flow from operations was \$893 million and free cash flow was \$832 million. The company had 135.3 million diluted weighted average shares outstanding at the end of the quarter.

The breakdown of revenue by reportable segments, end markets, major products, and regions, can be found within the shareholder letter and slides.

Turning to the balance sheet. KLA ended the quarter with \$4.5 billion in total cash, cash equivalents and marketable securities, debt at \$6.7 billion, and a flexible and attractive bond maturity profile supported by strong investment-grade ratings from all three major rating agencies.

Moving to our outlook. We believe our business is transitioning from a period of stabilization to a resumption of growth which began in our June quarter and we expect to continue through the remainder of calendar '24 and into 2025.

For calendar 2024, we remain encouraged by the improvement in our customers' revenue and profitability over the course of this year. This improvement will ultimately translate into new investment in capital equipment to support semiconductor growth over the medium-term. Our high-level outlook for the industry remains largely unchanged.

Our expectation is for the WFE market to be in the mid-\$90 billion range and that the second half of the calendar year will be stronger than the first half. While too early to be overly specific on expectations for calendar 2025, we do expect a year of growth due principally by growth in leading-edge investments in both Logic & Foundry and in Memory. Given KLA's business momentum, we are confident in our relative performance opportunities moving forward.

KLA's September quarter guidance is as follows. Total revenue is expected to be \$2.75 billion, plus or minus \$150 million. Foundry & Logic revenue from semiconductor customers is forecast to be approximately 80%, and Memory is expected to be approximately 20% of Semi Process Control systems revenue. Within Memory, DRAM is expected to be about 83% of the segment mix and NAND, the remaining 17%.

Gross margin is forecasted to be in a range of 61.5%, plus or minus 1 percentage point, as higher revenue volume is offset by weaker anticipated product mix.

Consistent with our comments last quarter, based on the current industry outlook, top-line growth expectations for calendar '24, higher forecasted growth in services, and expected system product mix, we are still modeling non-GAAP gross margins to remain relatively stable around the mid-61% range. Variability quarter-to-quarter is typically driven by product mix fluctuations.

Operating expenses are forecasted in the September quarter to be approximately \$565 million as we continue to make important R&D and scaling investments to support expected revenue growth. Looking ahead, we expect approximately \$10 million to \$15 million incremental growth in quarterly operating expenses for the remainder of calendar '24 and into 2025. This is supported by our revenue growth expectations and in line with our 40% to 50% incremental operating margin business model.

Other model assumptions for the September quarter include; other income and expense net of approximately \$34 million expense, GAAP diluted EPS is expected to be \$6.69, plus or minus \$0.60, and non-GAAP diluted EPS of \$7, plus or minus \$0.60. EPS guidance is based on a fully diluted share count of approximately 135 million shares.

In conclusion, our business has transitioned from a period of stabilization to one of growth. We remain optimistic that the indicators of improvement we have seen will continue throughout the remainder of calendar '24 and into 2025. KLA is focused on delivering a differentiated product portfolio that anticipates customers' technology roadmap requirements and drives our longer-term growth expectations.

With the KLA operating model guiding best-in-class execution, KLA continues to implement strategic objectives which are geared to drive our performance. KLA's focus on customer success, delivering innovative and differentiated solutions, and operational excellence is what drives industry-leading financial and free cash flow performance and allows us to return capital consistently.

Return of scaling and increasing complexity has solidified our confidence in the increasing importance of process control in enabling technology advancements. This is not just an improving time-to-results and process integration in fab ramp, but also an optimizing yield across a volume production environment with high semiconductor device design mix. This bodes well for KLA's long-term growth outlook as near-term industry demand trends are continuing to improve.

In alignment with this, KLA's business is improving, and the long-term secular trends driving semiconductor industry demand and investments in WFE remain compelling.

That concludes the prepared remarks. Let's begin the Q&A. Kevin?

Kevin Kessel {BIO 16743902 <GO>}

Thank you, Bren. Operator, we're ready for you to provide instructions and begin the Q&A session.

Questions And Answers

Operator

(Question And Answer)

Absolutely. (Operator Instructions). In the interest of time, we ask that you please limit yourself to one question and one follow-up. We will now take our first question from Vivek Arya with Bank of America Securities. Please go ahead. Your line is open.

Q - Vivek Arya {BIO 6781604 <GO>}

Thanks for taking my question. Rick, recently, TSMC said they expect the number of new tape outs for 2 nanometer in the first two years to be higher than what they saw for 3 nanometer and 5 nanometer in their first two years. And I think they've also heard a number of AI customers accelerate their roadmaps. I'm curious, what does this mean for KLA in terms of your outlook and kind of your share of the WFE wallet? And do you expect 2 nanometer to be a bigger factor for 2025 or for 2026?

A - Rick Wallace {BIO 3540990 <GO>}

Thanks, Vivek. Yes, we're definitely having those same conversations with our customers in terms of -- in the reticle world, of course, when there are more design starts, we see activity in the reticle market in terms of inspection for those reticles and then early launch. We're still early in n2. And in terms of your question, it's really going into '25 and then '26.

But the other thing we've seen is strengthening in n3 as well. So, I think both of those factors are being driven by some of the same drivers we talked about in the prepared remarks in terms of what we're seeing from AI driving that.

So, it does lead us to believe that our goals to increase our percent of WFE continue to be supported by the opportunities afforded by the additional process control challenges. So I think we're in pretty good shape with that. The other thing, of course, that's being driven in some of these advanced AI chips is larger die size, which, as you know, are a bigger challenge from a yield perspective and creates more incentive for more inspection, especially as these processes are brought online and ramped up.

So, mostly what we see is it's '25 leading into '26 in terms of when they reach HBM.

Bren, anything to add?

A - Bren Higgins {BIO 18108981 <GO>}

No, I think you covered most of it, Rick. I would say also one of the things that we're encouraged by that with the design start environment, given the high mix of parts, that our participation with our customers in terms of process control and monitoring that happens in production, tends to be at a higher level. So, they're trying to manage a robust design environment, but testing design rules in different ways. And that tends to be positive for continuous process control investment as they ramp up the node.

A - Rick Wallace {BIO 3540990 <GO>}

And remember, earlier in the year, we talked about a more positive tone we were seeing from our customers, and it was related to the early demand that they were starting to sense from their customers. And that's why we believed and that's how it's playing out that we would see strengthening our business. Maybe what's new, a little bit new, is the strength -- additional strength in n3, which wasn't as forecasted as the rollout of n2. So I think both of those are adding to our belief that we're in a good period for consistent acceleration and demand on the leading edge.

Q - Vivek Arya {BIO 6781604 <GO>}

Now on advanced packaging, you raised it from \$400 million to \$500 million. I wanted to clarify is that on an exiting full year basis or is that for the full year? So just that clarification. Then how would you break it out in terms of specialty and inspection metrology?

A - Bren Higgins {BIO 18108981 <GO>}

Yes. It's a good question. So that's our expectation for the business level in 2024. So it isn't a run rate necessarily, but what we expect to do which is significantly above what we did last year, which was close to the \$300 million. So a mid-60% growth rate with continuing momentum. I would think that we would continue at least through the first half of next year to see strong demand around packaging. So it's really an opportunity for -- certainly, we're seeing a lot of customer interest as we're seeing more and more focus on front-end capability, moving into the back-end. So it's an encouraging opportunity moving forward. I think the growth rate -- relative the growth rate at WFE will be higher over the next few years.

A - Rick Wallace {BIO 3540990 <GO>}

So one of the things that I'm sure you've heard is that one of the bottlenecks now for AI is, silicon is part of it, but the packaging co off [ph] is also part of what we're seeing. And we had customers for the first time in my career from the front-end start pulling us into packaging conversations because they want front-end capability, as Bren said, being, used in packaging to assist. So I think we talked about AI being a driver, it's really driving not just the silicon, but also packaging and also of course HBM for -- you need all three of those to fully execute on the plans people have.

So to your question on mix about, I'd say 60% of it is process control and then you have the remaining 40% is in our process -- specialty process business and also some exposure in terms of chemical process control as well.

But about 60% of it in the classic semiconductor process control part of our business.

Q - Vivek Arya {BIO 6781604 <GO>}

Thank you.

Operator

We'll take our next question from C. J. Muse with Cantor Fitzgerald. Please go ahead. Your line is open.

Q - C. J. Muse {BIO 6507553 <GO>}

Yes. Good afternoon and thank you for taking the question. In your prepared remarks, you talked about returning to growth for Memory in 2025. I know it's early, but would love to hear kind of your initial thoughts and maybe separating DRAM versus NAND, and then within DRAM, how you're seeing HBM conversion versus DDR5 and greenfields, and same for NAND layer counts versus any sort of expectations for greenfield in the 2025 timeframe.

A - Bren Higgins {BIO 18108981 <GO>}

2024 is really a transition year for Memory. We're not seeing a lot of growth overall, some in the DRAM front and some investment that's happened in China. So I think from a WFE point of view, 2025 is when we expect to see more growth there.

I think that -- you talked about the drivers, particularly around DRAM and HBM being drivers into next year. I think as we see our customers' business models improve, and we're certainly seeing the financial performance improve with pricing going up so significantly over the course of this year, but that will translate into investment as we move into next year.

I think on the flash market, we're less bullish into next year, although it's operating off a pretty low level. We'll see as we get closer to it, but right now, we do see a recovery in both parts of it, but certainly led by more DRAM than flash on an absolute WFE basis.

A - Rick Wallace {BIO 3540990 <GO>}

Yes. And I think much like some of what we saw in the early discussions of AI last year, where we didn't really see an increase in silicon because there was still utilization leverage that our customers have, some of that's in Memory, so we really don't see capacity starting to be added significantly until '25 and into '26 around HBM. So as Bren said, conversion, but then you don't get the capacity until later, and it's really HBM associated with DRAM before you see flash.

Q - C. J. Muse {BIO 6507553 <GO>}

Very helpful. And then as a follow-up, again, in the prepared remarks, you talked about confidence in your relative performance, and obviously no question around the Foundry & Logic side given the complexity and larger die size. But curious, as you see here increased Memory into the mix, is there enough share of wallet on the Logic & Foundry side to kind of mitigate that and allow that relative outperformance? Would love to hear the kind of puts and takes around that.

A - Bren Higgins {BIO 18108981 <GO>}

Yes, we feel pretty good, CJ. We are seeing relative performance this year, and this is a year that's still pretty heavy overall in what I'll call legacy levels of investments, particularly in China. So, as we start to move to leading edge and you see the entry ramp and also the initial investments; and then two, we would expect that, given our position, particularly in certain markets that we expect to scale and grow much faster than the overall, that we are pretty well-positioned into next year to see Foundry & Logic leading edge investment drive share of the overall spend.

We were talking about DRAM and HBM. HBM is a device that tends to -- first of all, you have advanced DRAM, you have EUV as a -- you get the introduction for use of EUV with those chips, but also as you start to put the chips together, it creates not only potential packaging opportunities moving forward, but you also have increased sampling to ensure that each of the DRAM die is functioning properly.

So, we've seen DRAM improve from a process control intensity point of view since the introduction of EUV, and so I would expect to see that continue as well as we move forward. So yes, I think we feel pretty good about the relative positioning of the company as we move into next year.

A - Rick Wallace {[BIO 3540990 <GO>](#)}

So, C. J., one of the things that we're seeing and we haven't seen in many, many years, if you remember part of the reason Memory had lower adoption was because of repair. And it seems that on leading DRAM repair is less effective than it's been. And so that is a factor that's driving some of our customers. Obviously, they're going to try to get through that. But that's -- they're expecting that they're going to have larger need for, a greater need for process control because of that.

And as Bren said, you also have -- because of EUV, you get the introduction of print check into memory and they have issues associated with pellicles. So -- I know it's very much into the weeds, but it drives more print checks. So the process control adoption is setting up to maybe be better than historical levels, at least for the advanced DRAM.

Q - C. J. Muse {[BIO 6507553 <GO>](#)}

Very helpful. Thank you.

Operator

And we'll take our next question from Harlan Sur with J.P. Morgan. Please go ahead.

Q - Harlan Sur {[BIO 6539622 <GO>](#)}

Yes. Good afternoon. Thanks for taking my question. As the team steps into the second half of this year, advanced technology node deployment is accelerating, obviously reflected in your strong guidance for the September quarter appears to be reflected in your confidence on the spending outlook this calendar year and growth into next. Near-term, does the team anticipate sequential growth in the December quarter? And then relative to, let's say, last earnings call, has your qualitative view on 2025 improved? I mean, I know, Rick, you mentioned incrementally better 3 nanometer investments. Any other indicators that are coming in better versus your prior expectations as you look into next year?

A - Bren Higgins {[BIO 18108981 <GO>](#)}

Yes. I'll start here first. We do expect sequential growth in the December quarter as we discussed last quarter when we talked about the March quarter being the bottom and an expectation of sequential growth through the year. So that's how we see things into the December quarter. We'll see how things play out in terms of what that looks like. I would say on the margin where we're incrementally more bullish, certainly when you look at the

funnel in terms of opportunities into next year, as we've seen that fill out, we have seen strength in the funnel to support our outlook into next year.

Q - Harlan Sur {[BIO 6539622 <GO>](#)}

Perfect. And then for my follow-up, and this is sort of a follow-on to what Rick was saying, but as you mentioned, Memory has always had a lower process control intensity, but there are dynamics that are pushing advanced logic like capabilities, right, into Memory. You mentioned more EUV adoption in DRAM, but like, for example, in HBM/DRAM, I mean, that's requiring this very advanced logic chip that sits at the bottom of the stack, right, that's using advanced CMOS logic technology.

In NAND, the periphery and controller chip, which is increasingly becoming more advanced logic, is being processed separately from the NAND stack, right, and then bonded together. And then this whole NAND sort of bonding technology process appears to be a nightmare from a manufacturability and yield perspective. But taking in totality, I mean, it does appear that these dynamics are driving memory process control intensity higher. I mean, any way to quantify, I know historically memory intensity has been about 10%, but if you look at all of these trends over the next few years, like, where can you see that memory intensity sort of trending to?

A - Bren Higgins {[BIO 18108981 <GO>](#)}

Yes. It's a good question, Harlan, and you did cover all the advancements that have us pretty excited in terms of opportunity moving forward. We have seen intensity improve over the last couple of years or so, but what we're waiting for to lean in, if you will, into that improvement is to see a full cycle in terms of investment by our customers.

And so I think as we look into next year, I think that where we are today is probably we're historically around 10%, that we're probably in excess of 11% today. And I think moving forward, if we can execute and deliver some of the products and capabilities we have in development here, I think it creates a number of opportunities covering the things you talked about and some of the other things we talked about in the last question or two.

A - Rick Wallace {[BIO 3540990 <GO>](#)}

Yes. And just contrast that from a few years ago, Harlan. If you go back and you think about 3D, 3D happens and our customers go backwards in lithography, right? DRAM is not really pushing design rules nor pushing advanced logic around the DRAM, and repair is still working. Like, you take all those factors off, and you get more into the expected growth and intensity that you get in logic or in the leading edge foundry for similar reasons.

So, we haven't fully modeled it because it's really hard. Customers are still trying to figure it out, but there's more characterization happening at the leading edge, leading us to believe that there's going to be more implementation. But until they've been through the cycles, they don't know. But it is, to Bren's point, why we feel more bullish.

In our engagement in those customers, they recognize an increased need for the process control capability. And then you just add on this bonding packaging question and that's a huge issue. And you know the value of HBM is so high also. So I think there's a lot of factors. In many ways, there are plenty of opportunities. Then the other part of that

answer is making sure that our solutions actually are going to work for those opportunities and we're still proving that out.

Q - Harlan Sur {BIO 6539622 <GO>}

Perfect. Thank you very much.

A - Rick Wallace {BIO 3540990 <GO>}

Thanks, Harlan.

Operator

We'll take our next question from Harlan Sur -- I apologize, from Atif Malik with Citi. Please go ahead. Your line is open.

Q - Atif Malik {BIO 15866921 <GO>}

Hi. Thank you for taking my question. The first one is for Rick on gate-all-around PDC intensity. Some of your peers have talked about 30% increase in metrology intensity at 2 nanometer gate-all-around or 3 nanometer, and the reticle inspection peers have talked about EUV map that's actually coming down to 20 from 30 from 3 nanometers. So when you put these two trends together, I'm curious how you're looking at your overall PDC intensity, and does that imply that you're gaining share on the reticle side?

A - Bren Higgins {BIO 18108981 <GO>}

So let me unpack that a little bit. I think that it is definitely the case for gate-all-around that it creates new integration challenges, and so we've seen, and we've talked about this for some time now, the adoption of Gen 4 to support that in a new configuration to be able to resolve the issues around that. So that's a driver. It's also a different approach than what customers have used because of the architecture change. There's more metrology intensity being used and largely an increase in the need to control voltage thresholds. So VT control becomes more important, so that's also part of it. You also just have film thickness measurement and management higher.

So overall, we would anticipate, but that's all kind of rolled into how we model the increasing process control intensity that we've outlined before. There's really nothing that's different than when we modeled that going forward toward our 2026 plan in terms of overall process control intensity and our share performance there.

And, Atif, if you look back to the last local high in KLA's share at WFE, it was when the industry transitioned from planar to finfet structures. So typically, architecture changes do drive -- because of just the change that's happening and the immaturity of the change, it does drive process control intensity to a higher level. So historically, it's proven. So Rick talked about not only inspection opportunities, but you're depositing in a lot more layers, 20% to 30% more critical layers on the metrology side. So we think in a number of areas, it's going to create opportunities.

And you still have the scaling dynamics. While litho layers are constant, those are still pretty challenging layers in terms of from a defect control point of view. So we're pretty bullish about the opportunities. I think it's feeding into a lot of what we've said here over in the last few questions in terms of our confidence moving into next year.

Q - Atif Malik {BIO 15866921 <GO>}

Great. And as a follow-up, Bren, the revenue guide is a bit wider than usual, plus or minus \$150 million versus \$125 million. Is that just a function of higher revenues, or is it something else?

A - Bren Higgins {BIO 18108981 <GO>}

Yes, a function of higher revenues. It's about 5% or so. At some point, as we move through the different quarters and revenue ranges, we start to make an adjustment just because of the higher levels. So there's nothing indicative in that other than the revenue is up 9% higher. We decided to -- it was 9% in June and then the guidance is another 7%. So that was prudent to raise the range.

You have to also remember that we do have integers that are quite large. If you think about our Broadband Plasma systems, they could be anywhere even plus \$20 million depending on the configuration. So tools moving in and out can have an impact in terms of our overall. So it's really more just the increasing revenue level and not related to anything else.

Q - Atif Malik {BIO 15866921 <GO>}

Thank you.

Operator

We'll take our next question from Chris Caso with Wolfe Research. Please go ahead.

Q - Chris Caso {BIO 4815032 <GO>}

Thank you. Good evening. I guess the first question is on the trends that you're seeing in mature node, and specifically, well, I guess China revenue in general. I think your prior comments were an expectation for China about flat in the second half. Can you talk about if that's still your view and what you're seeing outside of the leading edge?

A - Bren Higgins {BIO 18108981 <GO>}

Yes. It's still my view. You have the absolute dollar basis, right? And then with the second half growing relative to the first half, the percent will start to move in the second half versus the first half. But on an absolute dollar basis, as I said last quarter, it's about flat. Now, depending on what happens, you've got a lot of new projects, and RevRec could be an issue with newer customers. So where it shows up in terms of which quarter, I'm not going to comment on. But if you just take a step back and look at a half-to-half point of view, it's very similar to what I said last quarter.

Q - Chris Caso {BIO 4815032 <GO>}

Got it. And then following on, you've made some comments about the impact of high NA in the past. I mean, one, could you remind us a little bit about the impact of high NA on process control? And then secondly, with regard to timing for that there's been some noise around that about sort of implementation schedules. Anything different than you see on your side recognize that you folks would probably be a little bit earlier in seeing that given the timing of when you install process control?

A - Rick Wallace {BIO 3540990 <GO>}

Well, I think that the difference for high NA, the most obvious difference is you're going to continue a scaling. So you're going to get smaller defects and you're going to have new challenges associated with reticle qualification. And that takes form both in the mashup and requal and print check.

So all those factors continue, but it's really an extension of the trends that we've seen. And when you get high NA, you'll get also just the mix will shift. You'll have some layers initially of high NA, and then you'll have others that are going to be the 0.33. I think what we're hearing and seeing from customers is very consistent with what we believed in terms of when HBM happens. We don't see any acceleration in the timing. If anything, there's the potential for delay in terms of the HBM implementation. There will be characterization work happening before that.

But like all technology transitions, we see the existing technology often extends longer and has more upgrades and is more cost-effective, and that's what drives customers. Ultimately, it's about the economics. And so I think what will happen is if they can do that, then they'll bias toward extending the life of the 0.33 and the efficacy of that in the leading edge, and then high NA comes after. So nothing has really changed in our view. No acceleration, if anything, potential for some additional delay in the implementation.

Q - Chris Caso {BIO 4815032 <GO>}

Got it. Thank you.

Operator

We'll take our next question from Krish Sankar with TD Cowen. Please go ahead. Your line is open.

Q - Krish Sankar {BIO 16151788 <GO>}

Hi. Thanks for taking my question. I have two of them. The first one is actually on China. And Rick and Bren, thanks for the insight on the China revenues. I just kind of wanted to ask the question in a different way. One of your peers at ASM International today said that China mature nodes are in a digesting mode and calendar Q1 was the peak. I'm kind of curious, are you seeing that? And how do you think about China's sales into next year, especially given your long lead time? You might have a better insight than others. And also is that a factor in your gross margin down just a smidge in September quarter? And then I have a follow-up.

A - Bren Higgins {BIO 18108981 <GO>}

Yes, Krish, on China, it looks, like I said earlier, I think from a half-to-half point of view, it's relatively flat. So I don't see any real change. A lot of that's dependent on project timing given the level of greenfield activity that we're seeing there. As we move into next year, my early view on our business levels is more or less consistent with what we're seeing this year. I think when you look at the order funnel, what we have in backlog, what we have in deposits, it gives me some confidence in terms of what we'll see into next year. So I don't see a change. Obviously, the mix will be changing. There's infrastructure investments. It's obviously the Logic & Foundry, Memory, and so on. But from an overall level, it seems that it's pretty consistent year-to-year.

On gross margin, it -- while we outperformed in the June quarter, it was a little bit more than what we had initially modeled in the quarter. There's some adjustment that we're seeing in the September quarter from that. When you look at KLA gross margins, I talk a lot about product mix being the biggest driver of our gross margin in terms of variability quarter-to-quarter. It's not customer mix. It's not region mix. It is the portfolio mix. And not all the products in our portfolio carry the same margin profile, but also within the product families themselves, particularly when you're selling across multiple nodes with highly configurable systems, you can end up with varying gross margin profiles even within the same product families.

So it's really a function of that than it is anything else. And of course, the June quarter was stronger. I did put the comments, which are very consistent with last quarter in terms of expectations for this year in the mid-61% range. That's our expectation. And then moving forward, as we see the business continue to grow into '25, I think you'll see KLA's general incremental gross margin performance in and around that 65% incremental gross margin, plus or minus, as we move forward here. But in any given quarter, it could add a little bit of variability, but nothing that's different than what we usually see.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Got it. Very helpful, Bren. And then just to follow-up on the advanced packaging. Thanks to the color, like the \$500 million run rate from \$400 million like three months ago, is this really a function of the advanced packaging market growing? Are you like targeting your markets like macro inspection, which I feel like your peers or your competitors are doing better than that, are you like targeting those markets that's also helping some of this incremental revenue growth we did this year? I'm also kind of curious if you can extrapolate that into like panel level packaging, which is probably a few years out. Thank you.

A - Rick Wallace {[BIO 3540990 <GO>](#)}

Yes. I think there's no question that the package market is growing, and it's for all the reasons that you hear. I think that there's been acceleration, and again, I mentioned earlier in the call that some of the gaps, if you think about what's going on overall with AI, are packaging shortage related. So there's a lot of investment happening there. It's not just macro. It's what historically would have been considered pretty advanced lithography or inspection not that many years ago, because the design rules and the yield and what's at risk cost-wise is much higher.

Many years ago, when we bought a company and started doing its packaging, one of our customers asked, why are you even bothering, this is not a KLA market. What happened in the last year is now customers are saying, we need your capability that you have in the front end to be applied into some of our advanced packaging challenges and accelerated workshops with us talking about that, accelerated product roadmaps, and a lot of customer excitement and interest. So they're driving it very hard because they have shortages. And it's really across many of our products, both in our processing, but also in our process control. So it's been significant to see the growth and the speed of it. As we said, that's the annualized -- that's the annual number for calendar '24, but we'd expect the trend to continue toward growth in '25 and beyond.

A - Bren Higgins {[BIO 18108981 <GO>](#)}

Yes. Most of our growth has been more on the logic side. And I think one of the things we're excited about moving forward is that as you move in high bandwidth memory to hybrid bonding techniques, that you do increase the complexity of the process a fair amount. And as you know, the KLA position in the market is one of higher value offerings supporting more complexity. And so we're encouraged by what those opportunities might look like as we get out into the '26 and '27 timeframe.

Obviously, the industry's thinking a lot about what's next in terms of panel level packaging and so on, and so there's obviously R&D that goes into that, but so that's work that we will be doing in the company. And so I think that those opportunities, to Rick's point, are just showcasing, I think, what's the opportunities that are available in this part of the market over the next several years.

A - Rick Wallace {[BIO 3540990](#) <GO>}

Just one final point. Some of our customers -- our competitors are coming at this from a lower-end, and in many cases, the requirements are at the top-end of their capability. And so what we're hearing from our customers is, even though in some cases they're buying our systems, what are at the bottom-end of our capability, what they also want is the roadmap.

So I think they want the ability to buy that capability now, the expectation and the knowledge that they can upgrade it over time as their requirements continue, because you are now with that packaging. There's no question that roadmap is going to look more like a traditional scaling roadmap going forward.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Got it. So it's future-proofing. Thanks, Rick. Thanks, Bren. Very helpful. Thank you.

A - Rick Wallace {[BIO 3540990](#) <GO>}

Thanks, Krish.

Operator

We'll take our next question from Joe Quatrochi with Wells Fargo. Please go ahead.

Q - Joe Quatrochi {[BIO 18961101](#) <GO>}

Yes. Thanks for taking the question. I just wanted to kind of understand, I think looking at the guide for the September quarter and thinking about sequential growth into the December quarter, is it more fair to assume now our kind of thinking about low-double-digit, half-on-half growth rate for the calendar year relative to the high-single-digit last quarter that you talked about? And then kind of dovetailing that, I think last quarter you talked about process control systems mix as being closer to, like, 70% for foundry logic. Is that maybe a bit higher than that now?

A - Bren Higgins {[BIO 18108981](#) <GO>}

Yes. So on those fronts, so, again, I don't want to guide December, but I think high-single-digit, low-double-digit is probably the right way to think about our performance for the year overall. So I think that as we saw in June relative to the expectations we had a few months ago, we did better than we thought. September guide was higher -- certainly

higher than what consensus was. So I think that incremental optimism I talked about earlier in the funnel is revealing in some incremental growth here through the second half of the year.

When I look at the overall mix of business, I think that foundry logic is likely closer to 75%. So last year was 70% -- 70.30% on for approximate semiconductor process controls shipment or revenue to semiconductor customers. It's closer to 75%, 25% and 24%. But we'll see how it ends up.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Perfect. That's helpful. And then as a follow-up, the n2 strength you're talking about, I mean, given the lead times for Gen 4, Gen 5, I mean, is that going to kind of enter the model in terms of demand into next year? Is that how we should think about that, or some of those tools be pulled into 2024?

A - Bren Higgins {BIO 18108981 <GO>}

Well, we are seeing more capacity come online. Those are products where we've been supply-constrained, certainly around Gen 4. So, as new capacity comes online, I think it's one of the challenges for us, frankly, back in 2023 is we were limited. And as we move through this year, from a quarterly run rate point of view, we are getting incremental supply. And that, given the supply constraints and the level of demand, we are seeing growth there and would expect to see that continue into next year. So I'm fairly bullish on that part of the business growing faster than the overall market both this year and next year.

So I think that's one of those areas that I said earlier about our relative confidence in terms of share of the overall market is our presence in certain markets that are growing extremely fast relative to the overall, and certainly, high-end wafer inspection is one of those.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Thank you.

Operator

We'll take our next question from Toshiya Hari with Goldman Sachs. Please go ahead. Your line is open.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi. Thank you so much for taking the question. I wanted to double-click on China and what you're seeing in the region. You guys gave a lot of color in terms of what you're expecting for the overall geo. But by application, any standouts as you think about DRAM, foundry, other applications, or maybe by customer type, mask-making, wafer-making, et cetera, whether it be trends you saw in the first half or if you think about the flattish outlook into the second half, any standouts to the positive side or the negative side?

A - Bren Higgins {BIO 18108981 <GO>}

DRAM was strong in the first half and will be in the second half. I would expect that wafers probably a little weaker. I think on the reticle side, it's a little bit stronger, and

then foundry logic is stronger offsetting the weaker memory.

A - Rick Wallace {BIO 3540990 <GO>}

Some weakness in automotive.

Q - Toshiya Hari {BIO 6770302 <GO>}

Got it. Thank you. And then as a follow-up on your services business, you guys grew 14% year-over-year and you're outperforming your peers. And I think that's off of a higher base relative to your peers, too. So really good to see that. You talked about growing toward the high-end for the remainder of the year, if I'm not mistaken. Is that just simply utilization rates across our customer base improving, or is there more to it? And as a quick follow-up, is 12% to 14% still the right range?

A - Bren Higgins {BIO 18108981 <GO>}

Yes. So utilization rates are certainly a factor, so we've seen those steadily improve across all segments over the course of this year, so that's been a driver of incremental revenue. The other thing, and we had pretty good visibility to this, was the shipments that we, or the tools that we shipped in '21 and '22 have come off of warranty and gone into contract, and our contract renewal or attach rate is about 95%. So as those tools go off a warranty and into contract, that that's a driver. So we've had pretty good visibility into the service business.

We've had a headwind from FX, because you do have some of the service revenue denominated in local currency, particularly in Japan, so there's been some headwinds to the growth rate, frankly. But yes, we're trending towards that 14% sort of top part of the guidance range, and I think -- or the mark, the target we put out there. And I think as we go over the next couple of years, I feel like we're closer to the top-end of the range than the bottom-end of the range. Thank you, Toshiya.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you.

Operator

We'll take our next question from Joe Moore with Morgan Stanley. Please go ahead.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you. I also had a couple of China follow-ups. At one point, you had had a pretty sizable multi-national business in China. Is the business today mostly dominated by the China sovereigns, I assume?

A - Bren Higgins {BIO 18108981 <GO>}

Yes. Very, very little multi-national business in China.

Q - Joseph Moore {BIO 17644779 <GO>}

Okay. And then on the China sovereign side, I mean, having seen that business shift to more of a trailing-edge focus, what's happened to process control intensity? I mean, you

seem to be keeping pace with all of your competitors in terms of China exposure. So you seem to be doing quite well. I might have thought that the process control intensity would drop off, given the focus on lagging-edge technology.

A - Rick Wallace {BIO 3540990 <GO>}

Sure. It's definitely not as high as what you would see on leading-edge, but often in the case if you have small -- relatively size-wise small sites, then you're not really up against the percent -- against a huge number of wafer starts. So that's one factor. The other factor is some of the infrastructure that people are adding in terms of whether it's mashups [ph] or wafers. So in aggregate, that's why it's for us pretty much help serve.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you so much.

A - Bren Higgins {BIO 18108981 <GO>}

And, Joe, on the multi-national question, I mean, most of that business is service, right? We're not really -- there's no real equipment business that's happening with our multi-national customers there, but there is service activity supporting those fabs.

Q - Joseph Moore {BIO 17644779 <GO>}

Got it. Okay. Thank you so much.

Operator

And we'll take our next question from Srini Pajjuri. Please go ahead. Your line is open.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thank you. Hi, guys. A couple of longer -- actually one question about 2025 WFE. Rick, obviously, demand is improving across the board and China is stable. So that makes sense that WFE will grow next year. I'm just wondering in terms of -- in your assumptions, what sort of assumptions are you making as far as the chips money is concerned, because the check seems to be coming in, in second half. Just wondering if you're getting any explicit signals from your customers that that money is being spent in 2025.

A - Rick Wallace {BIO 3540990 <GO>}

Yes. Our forecasts are entirely independent of any chip back money. There's no dependence on that. If our customers receive that, it's going to be -- they're going to be getting it. What they tell us is to offset costs of building fabs in places that are more expensive. But it's not contributing to that. They're going to build capacity based on demand. The demand is going to be based on their market and the overall market, and then they'll buy equipment accordingly. But we're not counting on our customers to get chip back money to make our plans.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Okay. Thank you. And then more of a technology question, Rick, I time to time get questions about APMI for mask inspection. My understanding is that you guys were involved in that market and maybe exited at some point. I'm just curious, maybe I'm

wrong here, but I just want to get your thoughts on that as to how you're approaching that market and what your strategy for that market is.

A - Rick Wallace {[BIO 3540990 <GO>](#)}

Yes. We have not -- we were heavily involved in the reticle market. It's a market that really we started a company with and we're still involved in it heavily. The question on do you need actinic inspection for reticles has been a long debate that we've had. And our conclusion based on all the analysis that we've seen is the real market need for that is when high NA gets introduced, because at the existing technology nodes, there's really three ways to deal with the problems.

One is the challenges of reticle quality. One is the flagship product line that we've had for a long time, the 6xx and extensions to that. The other one is an ebeam product that we've talked about that's being used for very high resolution. And right now, it's the highest sensitivity reticle inspection tool still. And with our customers, we're still in the characterization of that tool that's the 8xx. And then the last one is printing onto wafers and verifying the goodness of the reticles. And that's what we're using -- basically, using the Gen 5 print check application for.

When it comes to investment in the actinic or the product that we've identified for high NA, we're pretty confident of our ability to intercept the high-volume manufacturing of high NA, which is still some years away.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Thank you. Very helpful.

Operator

We'll take our next question from Brian Chin with Stifel. Please go ahead.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Hi. Good afternoon. Thanks for letting us ask a few questions. Rick, you talked about inflections and sequential and year-over-year revenue growth. We're also seeing this in terms of bookings. And did your 12-month shippable RPO increase this quarter?

A - Rick Wallace {[BIO 3540990 <GO>](#)}

So we'll have all the specifics on that in the 10-K that we'll file here another week or two. It was pretty close to flat, down about \$70 million quarter-on-quarter. So pretty flattish compared to last quarter. And I think given what we're seeing in the funnel, I would expect it likely to probably over the next few quarters, we'll start to see it increase. But yes, just slightly down from last quarter.

Q - Brian Chin {[BIO 5763426 <GO>](#)}

Okay. That's helpful. Then I guess you've made a lot of disclosures in the past couple of calls about advanced packaging, and you saw even a step up on the year forecast over the past three months. Can you sort of ballpark how large the advanced packaging process control TAM is? You're probably not at your kind of typical, on average, 50%, 60% market share in process control overall. But maybe a sense of where the starting point is

now. And obviously, I'm sure you expect the market to grow and probably your share as well in the years ahead.

A - Rick Wallace {BIO 3540990 <GO>}

So I'll let Bren deal with the specifics. This is Rick. But it's not just process control for us. It's also process tools. Because of the SPTS acquisition we did about five years ago.

A - Bren Higgins {BIO 18108981 <GO>}

Yes. And look, we -- right now, our view is that this part of the market will grow faster than overall WFE. So if you look in the long run, as we expect WFE over the next few years, with capital intensities slowly rising to grow a little bit faster than semiconductor revenue that this part of the market will likely grow faster than that.

So that's the only quantifying we've done at this point. To Rick's point, we have a broad portfolio. We're seeing parts of it on the inspection side move up market. But we also sell a number of process tools. We also sell some chemical process control capability as well. So we're approaching this market in multiple ways. And like I said, I think that, as we talked about earlier, in the memory part of the market as it relates to HBM, we think we're better positioned relative to some of the technology transitions that are coming in terms of our share opportunity moving forward.

Q - Brian Chin {BIO 5763426 <GO>}

Great. Thank you.

Operator

We'll take our next question from Blayne Curtis with Jefferies. Please go ahead. Your line is open.

Q - Blayne Curtis {BIO 15302785 <GO>}

Thank you for taking my question. I had two. I want to ask you on the outlook for EPC. I think you were looking for growth last time. It might have come in a little bit below. I'm just curious, since we're going through all the calendar '24 forecasts, if you could update that one.

A - Bren Higgins {BIO 18108981 <GO>}

Yes. On specialty semiconductor, quarter-to-quarter, it dipped a little bit in the June quarter. But that's more of a timing issue. I think our expectation in that business is still some modest growth for the year. That business has a pretty diversified market exposure. And as power semi related to automotive has come down, you see packaging pick up. But there's a little bit of a dip in June. But it recovers and has a much stronger second half.

Flat panel business is where we announced our exit at that. And so we still have revenue in our plans for that as we exit that business and we're expecting the manufacturing likely in Q1 of 2025. On the PCB front, which is more consumer-centric, we still have -- that market's been relatively sluggish. It's improved a bit, but not much. And that's obviously tied much more to mobility. And so improvements into the future as we see more cell phone unit growth will drive that business.

Now, there was -- that business had a very strong '21 and into '22, a lot of capacity added during that that COVID period. And so the industry is still working off some of that capacity. Now, in that business, we do have service exposure. We have pretty good contract penetration in the PCB world. So while the systems business has been slower, there have been service contributions from a revenue point of view that we've had over the last year or so. But I would think that overall, the aggregate part of that business is up a few points year-to-year. I think it's we'll call it low to mid-single-digits.

Q - Blayne Curtis {BIO 15302785 <GO>}

Thanks. And then I just wanted to ask you, I know there's been a couple of questions on China. I heard from others that you're hearing less new names and I think the thought was maybe it would be more build out of facilities that were already in process. You talked about greenfields there. And obviously, there's another slug of money potentially coming in. So as you look out a year or more, I'm just kind of curious how you think about that frenzy we had before of all these new names that you never heard of before, creating companies, are you seeing any view that that might come back?

A - Bren Higgins {BIO 18108981 <GO>}

Well, there's still some new business from new customers. The other thing that's happening in China is you do see investment from our mature legacy customers in China. And so where we've been engaged in supporting those customers for many, many years, and their businesses are typically driven by supplying for specific markets and in response to expectations of demand moving forward. So you do have those customers also investing more here, I think, moving forward. And then you do move into second phases with certain ones, but there's also some new ones too. So it's a pretty broad mix as I look at the remainder of this year and into next year.

Q - Blayne Curtis {BIO 15302785 <GO>}

Got you. Thank you.

Operator

And we'll take our last question today from Tim Arcuri with UBS. Please go ahead.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks. I had two. So, Bren, my question on China. So there's been some recent news, obviously the U.S. Department of Commerce is trying to use the foreign direct product rule against some of the non-U.S. suppliers. And it would kind of suggest that maybe there's some forthcoming news that there's a reason why they would want to use the FDPR, like maybe some entity list additions. So how do you think about that? I know it's kind of hard for you to even handicap your guidance for that. And I assume that you're not accounting for that at all because you don't even know like whether that's going to come, but how do you kind of think about that and what the likelihood is and how do you even handicap for it?

A - Bren Higgins {BIO 18108981 <GO>}

Yes, Tim, I mean, there's been a lot of news for some time and so you're right, I can't participate here in speculation and hypotheticals about what could or could not happen.

So we will continue to assess, we work very closely with the U.S. government. Of course, we do everything we need to do in the company to be in compliance. We'll follow the laws we do and we'll support our customers as well within the confines of whatever the regulations are. So if and when something happens, we'll assess the impact and let you know what the impact is in sort of the near-term and what might be over the line.

A - Rick Wallace {BIO 3540990 <GO>}

Well, one of the things we talked about a lot, Tim, and I know you know this, is because of where we are with the AI drivers, we're seeing an acceleration at the leading edge. So if we think about what's really going to propel our business going forward, it's not so much the legacy, which has been maybe overrepresented on historical levels last few years, but it's the stuff we're seeing right now, especially at the leading edge in terms of leading edge foundry. And that's been driving a lot of our business and our expectations around what we're seeing in n3 and n2, plus the packaging, plus we're seeing in HBM. So if I think about '25, '26, those are going to be the things that drive KLA, drive our performance, and should drive good opportunities for our investors.

A - Bren Higgins {BIO 18108981 <GO>}

If you go back to 2022, 75% plus of our business was what we'll call leading edge, advanced design rules 20 nanometer or below. So with the expectation of a ramp into next year from our more leading edge customers on the logic and memory side we think that that's going to drive the performance of our business as we move over the course of next year and beyond.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay. And then last thing. So book-to-bill, based on the RPO, book-to-bill was just a touch below 1. So this is the seventh quarter that it's been below 1. And it sounds like you think book-to-bill is going to start to drift up actually, that we're going to finally start to be above 1. So is there some dynamic where it seems like there was this massive slug of bookings that happened back in mid '22 and we've been kind of shipping off of that ever since. And are we kind of getting to a point now where you think that customers are kind of coming back in and they're -- and we could begin to see book-to-bill start to drift above 1 where we can start to not just be shipping off of this big slug of orders from the middle of 2022?

A - Bren Higgins {BIO 18108981 <GO>}

Yes. I think based on the funnel that we will see, yes, I think the order profile looks pretty good. Now orders are not always indicative of what's going on given the number of customers and the size of the orders. And then when it comes to KLA, even the size of the ASPs of certain types of products. But all that being said, yes, it went from about 13 billion at its peak in terms of RPO down to just sub 10 billion today. So it's still quite large, but I would expect that moving forward, that at least over the next couple of quarters, it looks that the order profile looks decent. So yes, I think that's a reasonable way to see it. Thank you, Tim.

Operator

I'll now turn the floor back over to Kevin Kessel for any additional or closing remarks.

A - Kevin Kessel {BIO 16743902 <GO>}

All right. On behalf of KLA, we appreciate your time and your interest. We'll be seeing many of you throughout this quarter at different conferences. If you have any follow-up questions, please do reach out to the KLA Investor Relations team. With that, I'll turn it back to the operator to conclude the call.

Operator

This concludes the KLA Corporation June quarter 2024 earnings call and webcast. Please disconnect your line at this time and have a wonderful day.

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